

Childcare Affordability Across U.S. Counties

A Data Storytelling Project

Shaghayegh Malekshahi

Bellevue University

DSC640 – Data Presentation & Visualization

Introduction

Childcare affordability has become an increasingly significant economic and social issue in the United States. For many families, childcare represents one of the largest household expenses, particularly during early childhood when full-time care is often required. High childcare costs can influence workforce participation, family financial stability, and long-term economic opportunity. As policymakers and workforce development organizations attempt to address these challenges, reliable data and clear communication of affordability patterns become essential.

This project examines childcare affordability across U.S. counties using data from the National Database of Childcare Prices. The goal of the project is to identify patterns in childcare costs, explore how those costs relate to household income, and communicate affordability challenges through effective data storytelling. By combining quantitative analysis with visual communication mediums, the project demonstrates how data can support evidence-based policy discussions and improve understanding of childcare affordability challenges.

Dataset Description

The analysis uses the National Database of Childcare Prices, a dataset published by the U.S. Department of Labor Women's Bureau. This dataset provides county-level estimates of childcare prices across the United States and includes information about childcare costs for different age groups and provider types.

Key variables used in the analysis include weekly childcare prices for infants, toddlers, and preschool children, as well as price estimates for both center-based childcare providers and family childcare providers. The dataset also includes county-level median household

income values, which allow childcare costs to be evaluated relative to household financial capacity.

To maintain consistency and interpretability throughout the analysis, the project focuses on data from the year 2018, which represents the most recent year available in the dataset version used for the project. Because the data is aggregated at the county level and released by a federal agency, it provides a credible and reliable foundation for analyzing broad affordability patterns without including personally identifiable information.

Analytical Approach

The primary goal of the analysis is to evaluate childcare affordability rather than simply comparing childcare prices. To accomplish this, weekly infant childcare prices were converted into annual childcare cost estimates by multiplying weekly prices by 52 weeks. These annual childcare costs were then compared with county-level median household income values.

An affordability ratio was calculated to measure the share of household income required to pay for infant childcare. This ratio provides a standardized way to compare affordability across counties with different income levels.

Several data preparation steps were conducted before performing the analysis. The dataset was filtered to include only observations from the year 2018, and only variables relevant to childcare costs and income were retained. Records with missing values in core variables were removed, and observations containing non-positive values for childcare prices or income were excluded to prevent invalid affordability calculations.

The analysis then explored patterns across several dimensions, including childcare age groups, childcare provider types, geographic variation across counties and states, and the relationship between childcare prices and household income.

Key Findings

The analysis reveals several important insights regarding childcare affordability across the United States.

First, infant childcare consistently represents the highest-cost category across all childcare age groups. Although childcare costs decline somewhat for toddlers and preschoolers, infant care remains the most expensive stage for families requiring full-time

childcare. This pattern likely reflects the greater staffing requirements and regulatory standards associated with caring for infants.

Second, center-based childcare is consistently more expensive than family childcare providers across all age groups. This structural pricing difference highlights the role of provider type in determining childcare affordability.

Third, the analysis shows that affordability burdens vary widely across counties. Some counties experience relatively moderate childcare costs relative to income, while others face significantly higher affordability pressures. This variation indicates that childcare affordability challenges are not evenly distributed geographically.

Fourth, when county-level affordability data is aggregated at the state level, certain states exhibit substantially higher affordability burdens than others. These geographic patterns suggest that regional economic conditions, childcare supply, and policy environments may influence childcare affordability.

Finally, the relationship between household income and childcare costs reveals that higher income levels do not always eliminate affordability challenges. Some higher-income counties still experience high childcare prices, suggesting that regional childcare markets and supply constraints play a major role in determining costs.

Communication Mediums

To effectively communicate the findings of the analysis to different audiences, the project uses three communication mediums.

The first medium is a policy-focused presentation, designed for policymakers and workforce development stakeholders. This medium emphasizes clear visual comparisons and key affordability patterns to help decision-makers quickly understand where childcare costs place the greatest burden on families.

The second medium is an interactive dashboard concept designed for analysts and program managers who may need to explore childcare affordability patterns more deeply. A dashboard would allow users to filter the data by geography, childcare provider type, age group, and affordability thresholds in order to identify regions where childcare costs present the greatest challenges.

The third medium is a public-facing infographic, which is designed to communicate key findings to a broader audience, including parents, advocacy groups, and the general public.

Infographics emphasize accessibility and simplicity by presenting key insights in a visually engaging format.

Using multiple mediums allows the same dataset to be communicated effectively to audiences with different levels of technical expertise and different decision-making needs.

Design Decisions

The visual design of the project prioritizes clarity and accessibility. Because the intended audience includes policymakers and non-technical stakeholders, visualizations were selected based on their ability to communicate comparisons and distributions clearly.

Bar charts were used to compare childcare costs across age groups and childcare provider types because they allow straightforward comparisons between categories. Histograms were used to illustrate the distribution of affordability ratios and provider cost gaps, highlighting variation across counties. A scatter plot was used to explore the relationship between household income and childcare costs.

Consistent colors, clear labeling, and balanced visual spacing were used throughout the visualizations in order to reduce cognitive load and make the findings easier to interpret.

Ethical Considerations

Several ethical considerations were addressed during the preparation and analysis of the dataset. Data preparation included filtering the dataset to the year 2018, selecting relevant variables, removing records with missing values in key fields, and excluding observations with non-positive childcare prices or income values. While these steps improve analytical clarity, they can introduce bias if excluded records differ systematically from those included in the analysis.

Another ethical consideration involves the interpretation of results. The visualizations presented in this project show associations and patterns in the data but do not establish causal relationships. Care was taken to avoid overstating conclusions or implying that any single factor directly causes childcare affordability challenges.

The dataset used in the analysis is publicly available and published by the **U.S.**

Department of Labor Women's Bureau, which provides documentation describing data collection methods and imputation procedures. Because the dataset contains only aggregated county-level data and no personally identifiable information, there are no privacy concerns associated with its use in this academic project.

Limitations

Although the dataset provides valuable insights into national childcare affordability patterns, several limitations should be considered. Childcare prices in the dataset represent county-level estimates and may not reflect the exact costs experienced by individual families. Additionally, median household income is used as a proxy for household financial capacity, but income variation within counties may produce different affordability experiences for individual households.

Despite these limitations, the dataset provides an informative overview of how childcare costs vary geographically and how those costs relate to income levels.

Policy Implications

The findings of this analysis suggest several potential policy considerations for addressing childcare affordability challenges across the United States. First, the high cost of infant childcare highlights the importance of targeted support for families during the earliest stages of childcare demand. Programs such as childcare subsidies, tax credits, or employer-supported childcare initiatives may help reduce the financial burden faced by families with infants.

Second, the consistent cost differences between center-based childcare and family childcare providers suggest that expanding access to diverse childcare options may help improve affordability. Policies that support the development and sustainability of family childcare providers could increase the availability of lower-cost alternatives for families.

Third, the geographic variation in affordability burdens indicates that childcare challenges are not evenly distributed across regions. Policymakers may benefit from using localized data to identify high-burden areas and design targeted interventions rather than applying uniform national policies.

Finally, the relationship between childcare costs and household income suggests that affordability challenges are influenced not only by income levels but also by regional childcare market conditions. Addressing childcare affordability therefore requires a combination of economic support for families and policies that improve childcare supply and accessibility.

Conclusion

Childcare affordability represents a significant financial challenge for families across the United States, particularly for households with infants requiring full-time care. The analysis demonstrates that childcare costs vary widely across counties and states and that affordability burdens are not evenly distributed geographically.

By combining structured data analysis with multiple communication mediums, this project illustrates how data storytelling can transform complex datasets into clear and actionable insights. Understanding where childcare affordability challenges are most severe can support more targeted policy discussions and contribute to efforts aimed at improving childcare accessibility and supporting family economic stability.